Lab Book Layout — Experiment 2: Gravitational Acceleration

George · how to set out pages 5–10, following the same layout as Ryan's lab book

How to read this guide

- **Blue text** = write this in your lab book by hand (headings, calculations, results).
- **Grey boxes** = the points to cover. Write these sentences in your own words.
- **Orange dashed boxes** = cut out that page of the print pack and glue it in there.

What to print

Print Lab_Book_Print_Pack_George.pdf (6 pages) and cut along the dashed boxes.

Print pack page	Item	Glue onto lab book
1	Example of tracking the ball (Run 3)	Page 6
2	Recorded data for Run 3 (full table)	Page 7
3	Velocity vs time graph, Run 3	Page 8
4	Value of g for each run (or copy by hand)	Page 9
5	All 9 runs combined graph (extra)	Page 10
6	Air resistance check table (extra)	Page 10

Also save both Excel files (Experiment2_Gravity_George.xlsx and Experiment2_Data_Tables_George.xlsx) in your class computer folder for this lab — the lab book refers to them for the other 8 runs.

LAB BOOK — PAGE 5 Purpose · Apparatus · Procedure (group page, already written)

This page is already done. Make two corrections:

- Purpose, the equation: change $h = v_0 + \frac{1}{2}gt^2$ to **$h = v_0t + \frac{1}{2}gt^2$**
- Procedure step 2: cross out "240 fps in slow-mo mode" and write **120 fps (normal video)**

LAB BOOK — PAGE 6 Data Analysis

Data Analysis:

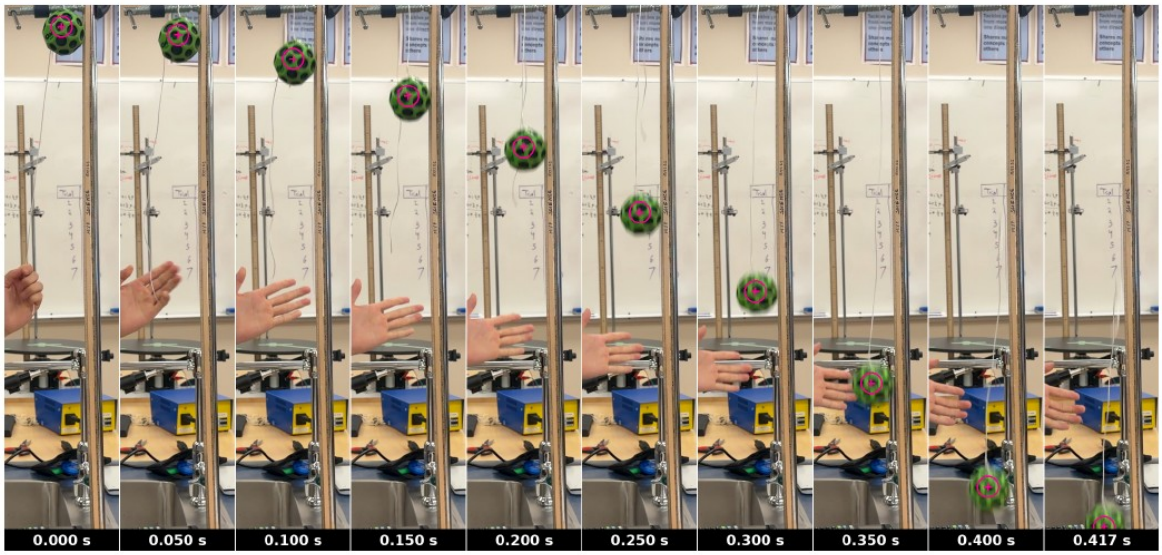
Write 2–3 sentences in your own words, including:

- The videos were filmed at 120 frames per second, so frames are 0.0083 s apart.
- The position of the ball was measured in every frame by tracking its centre with a video-tracking program (be honest that it was not Tracker).
- The scale came from the drop height: 82.5 cm from the bottom of the ball to the base, measured with a metre stick.
- Time t = 0 is the last frame before the ball starts to fall.

✂️ **GLUE IN: print pack page 1 — Example of tracking the ball (Run 3)**

Example of tracking the ball (Run 3, IMG_3051)

Every sixth frame of the fall (120 fps). The pink circle marks the centre of the ball measured in that frame; the time under each frame is the time since the ball was released.



↑ label it: "Example of tracking (Run 3)"

LAB BOOK — PAGE 7 Recorded data + example calculations

✂️ **GLUE IN: print pack page 2 — Run 3 table**

Recorded data for Run 3 (IMG_3051)

	Time (s)	Height (m)	Smoothed height (m)	Average velocity (m/s)
t_1	0.000	0.825		
t_2	0.008	0.824	0.824	−0.156
t_3	0.017	0.823	0.823	−0.233
t_4	0.025	0.821	0.821	−0.302
t_5	0.033	0.819	0.818	−0.399
t_6	0.042	0.815	0.815	−0.481
t_7	0.050	0.811	0.811	−0.551
t_8	0.058	0.807	0.806	−0.629
t_9	0.067	0.801	0.801	−0.703
t_{10}	0.075	0.795	0.795	−0.794
t_{11}	0.083	0.789	0.789	−0.910
t_{12}	0.092	0.781	0.781	−0.917
t_{13}	0.100	0.773	0.773	−1.014
t_{14}	0.108	0.766	0.765	−1.034
t_{15}	0.117	0.756	0.756	−1.235
t_{16}	0.125	0.747	0.746	−1.274
t_{17}	0.133	0.735	0.735	−1.380
t_{18}	0.142	0.724	0.724	−1.393
t_{19}	0.150	0.712	0.712	−1.467
t_{20}	0.158	0.700	0.700	−1.543
t_{21}	0.167	0.688	0.687	−1.630
t_{22}	0.175	0.674	0.674	−1.683
t_{23}	0.183	0.660	0.660	−1.762
t_{24}	0.192	0.645	0.645	−1.824
t_{25}	0.200	0.630	0.630	−1.955
t_{26}	0.208	0.614	0.613	−1.977
t_{27}	0.217	0.597	0.597	−2.068
t_{28}	0.225	0.580	0.580	−2.116
t_{29}	0.233	0.562	0.562	−2.235
t_{30}	0.242	0.544	0.543	−2.336
t_{31}	0.250	0.524	0.524	−2.400
t_{32}	0.258	0.504	0.504	−2.446
t_{33}	0.267	0.484	0.484	−2.503
t_{34}	0.275	0.463	0.463	−2.589
t_{35}	0.283	0.441	0.441	−2.688
t_{36}	0.292	0.419	0.419	−2.769
t_{37}	0.300	0.396	0.396	−2.831
t_{38}	0.308	0.372	0.372	−2.893
t_{39}	0.317	0.348	0.348	−2.948
t_{40}	0.325	0.324	0.323	−2.955
t_{41}	0.333	0.298	0.299	−3.048
t_{42}	0.342	0.274	0.273	−3.073
t_{43}	0.350	0.247	0.248	−3.194
t_{44}	0.358	0.222	0.221	−3.199
t_{45}	0.367	0.194	0.194	−3.489
t_{46}	0.375	0.167	0.165	−3.446
t_{47}	0.383	0.134	0.137	−3.517
t_{48}	0.392	0.108	0.107	−3.492
t_{49}	0.400	0.080	0.078	−3.726
t_{50}	0.408	0.047	0.047	
t_{51}	0.417	0.015		

Recorded Data for Run #3

Example of smoothed height for t_2

$h_{s2} = (h_1 + h_2 + h_3) / 3$

$= (0.82496 + 0.82438 + 0.82257) / 3$

h_{s2} = 0.82397 m

No h_s at t_1 or t_{51} because h_0 and h_{52} don't exist.

Example of average velocity at t_2

$v_2 = (h_{s3} − h_{s2}) / (t_3 − t_2)$

$= (0.82267 − 0.82397) / (0.016667 − 0.008333)$

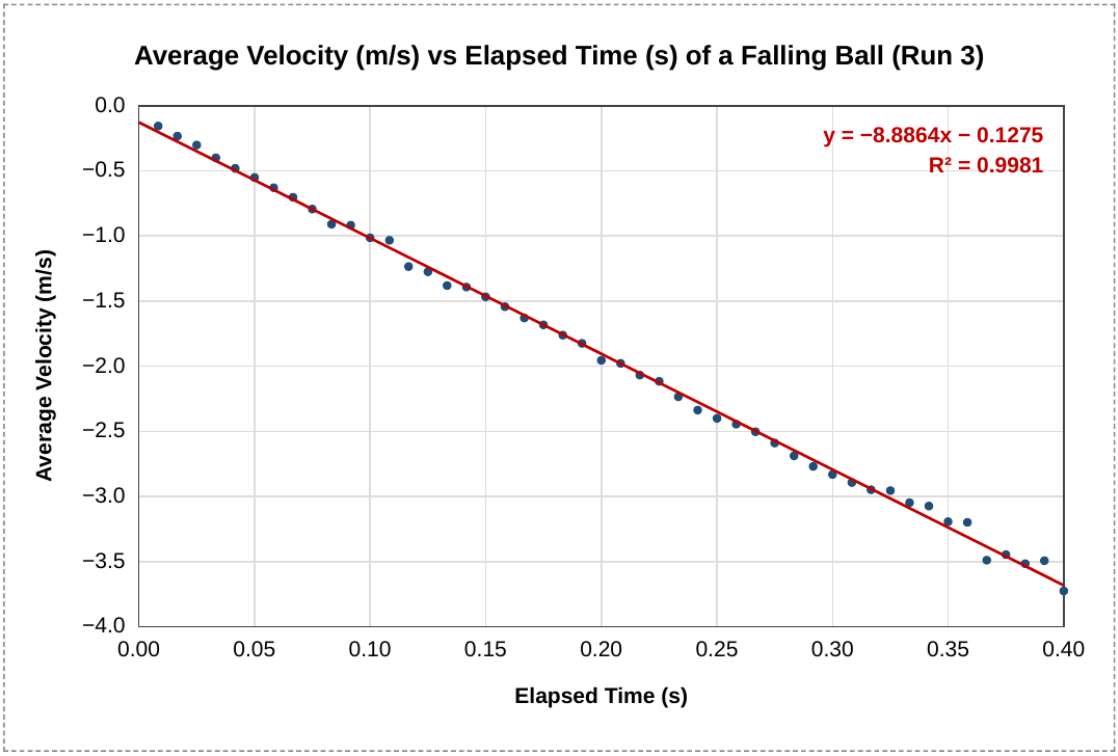
v_2 = −0.156 m/s

↑ *extra decimal places from the spreadsheet; with the 3-decimal table values the rounding gives the wrong answer.*

No velocity at t_1, t_{50} or t_{51} because h_s is missing there.

LAB BOOK — PAGE 8 Graph + determining g

✂️ **GLUE IN: print pack page 3 — Velocity vs time graph, Run 3**



Data for the other 8 runs is in the Excel files (Experiment2_Gravity_George.xlsx) in the computer folder.

Determining constant g:

Within the velocity–time graph above, −g is given by the slope of the trendline.

$y = -8.8864x - 0.1275$

$-g = -8.8864$

$\therefore g = 8.8864 \text{ m/s}^2$

LAB BOOK — PAGE 9 Results table · mean · uncertainty · conclusion

Copy this table by hand (or glue in print pack page 4):

Run #	Trendline Equation	R² value	Value of g (m/s²)
1	y = −8.6225x − 0.2339	0.9901	8.6225
2	y = −8.7772x − 0.1657	0.9980	8.7772
3	y = −8.8864x − 0.1275	0.9981	8.8864
4	y = −8.6806x − 0.1454	0.9971	8.6806
5	y = −9.0588x − 0.1516	0.9985	9.0588
6	y = −9.1246x − 0.0827	0.9982	9.1246
7	y = −8.8489x − 0.2167	0.9982	8.8489
8	y = −8.9518x − 0.1558	0.9975	8.9518
9	y = −9.0938x − 0.2318	0.9983	9.0938

Mean Value of g

$\bar{x} = \Sigma x_i / n$

= (8.6225 + 8.7772 + 8.8864 + 8.6806 + 9.0588 + 9.1246 + 8.8489 + 8.9518 + 9.0938) / 9

= 80.0446 / 9 = 8.8938

g ≈ 8.89 m/s²

Using TI-84 (1-Var Stats): Sx = 0.180 m/s²

Uncertainty: write in your own words, including:

- Air resistance is the main error, and you have evidence: g from the first half of each fall averaged 9.22 m/s² but the second half only 8.42 m/s² (light ball with holes).
- The ball was let go by hand, somewhere within one frame (8 ms) of t = 0.
- The drop height was measured with a metre stick (±0.5 cm of 82.5 cm, about 0.6%).
- One video (IMG_3041) was a duplicate of IMG_3055, so 9 runs were used, not 10.

Percent error = | Experimental − Theoretical | / Theoretical × 100

= | 8.89 − 9.81 | / 9.81 × 100 = 9.34 %

CONCLUSION

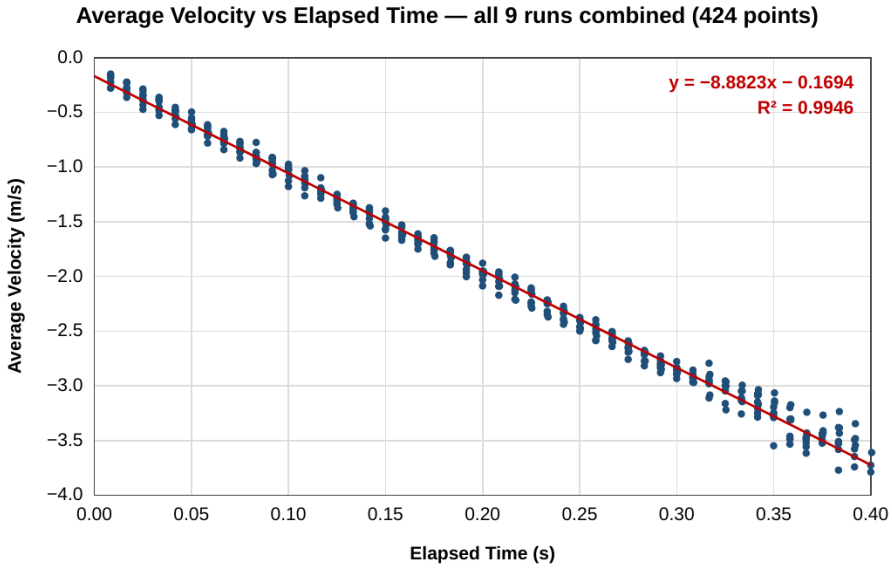
g = 8.89 m/s², Sx = ±0.18 m/s², percent error = 9.34 %

or rounded: 8.9 ± 0.2 m/s², 9 %

LAB BOOK — PAGE 10 (optional)

Extension: all runs combined · air resistance · improvements

✂ **GLUE IN: print pack page 5 — All 9 runs combined graph**



One straight line through every velocity point from all 9 runs: g = 8.882 ± 0.032 m/s² (standard error of the slope).

One trendline through all 424 points: g = 8.882 ± 0.032 m/s². Combining runs shrinks the random error, but not the error from air resistance.

✂ **GLUE IN: print pack page 6 — Air resistance check**

Air resistance check

g from the first half and the second half of each fall (slope of each half of the velocity–time data).

Run #	g, first half (m/s²)	g, second half (m/s²)
1	9.34	7.11
2	9.13	8.19
3	9.22	8.53
4	9.11	8.07
5	9.15	8.94
6	9.43	9.10
7	9.21	8.33
8	9.22	8.51
9	9.17	8.99
Mean	**9.22**	**8.42**

Improvements: in your own words, e.g.

- A heavier, solid ball (golf or steel) so air resistance matters less.
- A cleaner release (cutting the string, or an electromagnet) so t = 0 is exact.
- A higher frame rate (240 fps) for more points and better timing.
- A metre stick turned to face the camera, as a second check on the scale.